

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	<u>14M</u> TLO 1.1 Describe functions of an Operating System. TLO 1.2 Explain different services of Operating System. TLO 1.3 Explain use of system call of Operating System. TLO 1.4 Explain activities of Operating System in concern with their components.	Unit - I Operating System services and components 1.1 Operating System: concept, functions 1.2 Different types of Operating System: Batch Operating System, Multi-programmed, Time Shared Operating System, Multiprocessor System, Distributed System, Real Time System, Mobile OS (Android OS) 1.3 Command line based Operating System: DOS, UNIX GUI based Operating System: WINDOWS, LINUX, MaC OS 1.4 Different Services of Operating System, System Calls: Concept, types of system calls 1.5 Operating System Components: Process Management, Main Memory Management, File Management, IO Management, Secondary Storage Management	Presentations Lecture Using Chalk-Board

Operating System ? (Actual) (Need CPU) Concept

It is a set of programs which controls all the computer resources and provide environment in which user can develop application programs.

An Operating System is a software act as interface between end-users and hardware. It creates user-friendly environment.

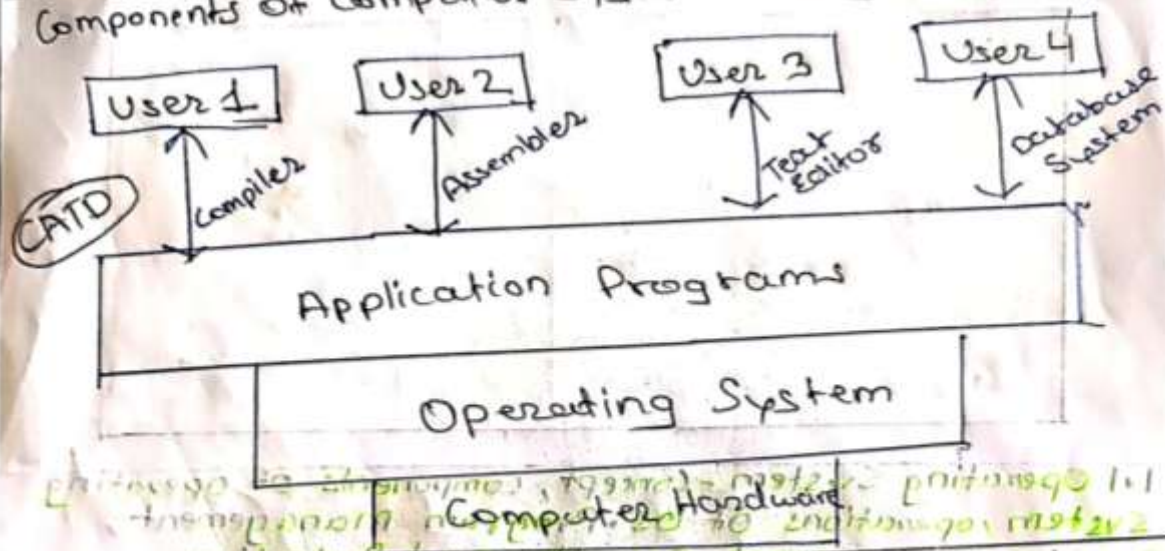
If there is no operating system user can't interact with the hardware. If he wanted to do without O.S the he have to write program for each and every thing.

Components of Computer System

A Computer System is a collection of hardware and software components.

Hardware refers to physical computing equipment.
Software refers to the programs written to provide services to users of the system.

Components of Computer System Diagram



Computer system is roughly divided in 4 parts.

① Hardware

Hardware is a physical part of machine which provides basic computing resources.

Examples:- Central Processing Unit (CPU)

- Input devices (Keyboard, mouse, microphone, web camera, and scanner etc)

- Output devices (Printer, monitor and Plotter etc..)

- Networking devices (LAN card, Router, Hub, Switch, modem etc) and other devices like motherboard, DVD writers etc..

② Application Programs

It is a software that performs specific functions for a user.

They define the ways in which the system resources are used to solve the computing problems of the users.

Examples are Word Processors, Spreadsheets, Compilers, web Browsers, Databases, Video Games etc...

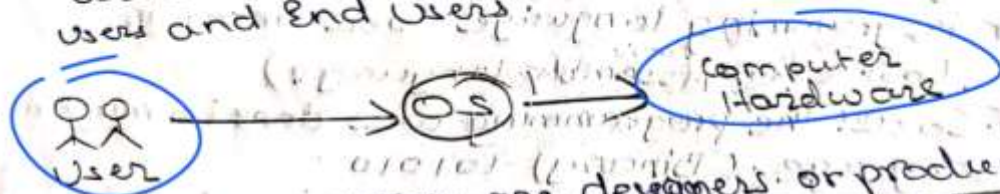
③ Operating System

Operating system is system software that manages a computer resources among various application programs.

Examples are Microsoft Windows - 7, Apple macOS, Ubuntu, Linux etc.

④ Users

User can be categorized into Programmer, Operational users and End Users.



- Programmers: They are designers or producers of software developed for themselves or others.

- Operational Users: They are the users responsible for the installation of software, proper maintenance of software, making it available for users.

- End Users: are the actual users of the applications and perform the specific tasks as per defined.

OPERATIONS/FUNCTIONS OF OPERATING SYSTEM

Q] List any four functions of operating system.

(Any four functions - 1 mark each)

Ans: The major functions of an operating system are:

1. Process Management – A program in execution is called as Process. OS is responsible for creating, deleting, suspending and resuming the processes.

2. Memory Management – OS is responsible for allocation and deallocation of memory to the system programs as well as other programs and data.

3. File Management – An OS deals with the **storage, creation and deletion** of file on various storage devices. It also allows all files to be easily **changed and modified** through the use of text editors.

4. Device / Resource Management – Managing devices and resources and allowing the users to share the resources such as **CPU time, main memory, secondary storage and input output devices.**

5. Security and Protection – Securing the system against possible **unauthorized access** to data or any other entity by means of passwords and similar other techniques.

6. Job Accounting- Keeping track of **time and resources** used by **various jobs and users.**

7. Communications – Providing **interface for the user to communicate** with each other in the same or other system by sharing or transferring data.

8. Booting the System and getting it ready to work.

1.2 Different types of Operating Systems:

As we know an operating system is mediator between the user and hardware.

An Operating system is essential software that manages a computer resources like files, processes and memory. It acts as Bridge between the user and machine, making it easier to interact with computer.

Types of operating System

1 → Batch Operating System

2 → Multiprogrammed Operating System

4 → Time shared OS → Multitasking

5 → Multiprocessor Systems

6 → Distributed Systems

7 → Real time Systems

8 → Mobile OS

↙
Android

↘
iOS

Batch Operating System

The users of Batch operating system do not interact with computer directly.

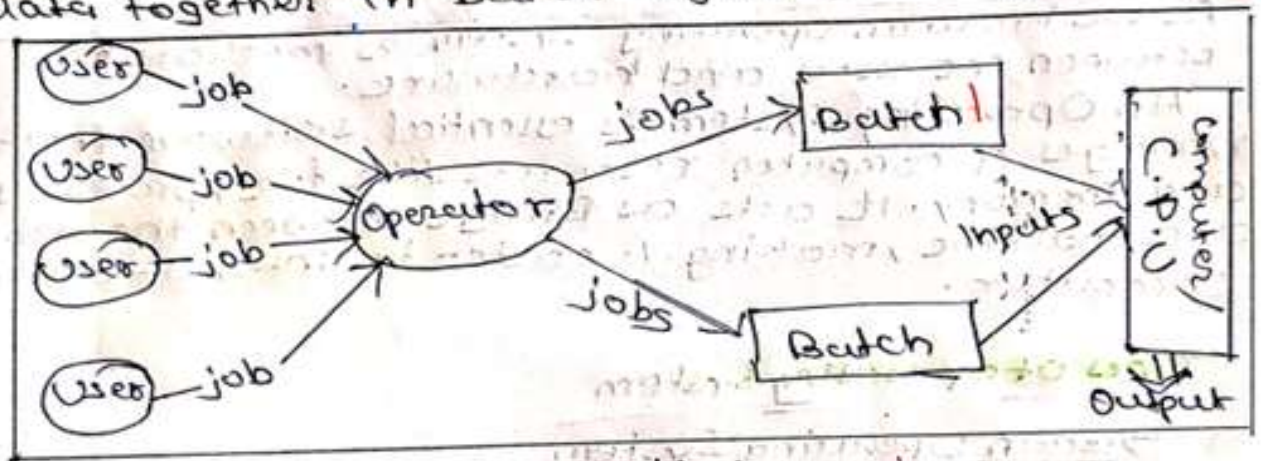
Each user prepares his job on offline device like punch cards and submit it to the computer operator (person).

The Batch operating system will work to submit similar kinds of jobs together.

Job: Job is collection of program, I/P data and control instruction.

In Batch operating system, a batch is a sequence of user jobs.

Here operating system collects program and data together in batch before processing starts.



It follows following activities

- ① Operating system defines a job which has predefined sequence of command, programs and data in single unit.
- ② Operating system keeps a no. of jobs in memory and executes them without any manual information.
- ③ Jobs are processed in order of submission i.e. First come first served.
- ④ After when jobs are executed the operator separates the similar kind of jobs in batch.
- ⑤ As an input Batches are given to Computer CPU it executes it in FIFO and process it one batch at a time & provide desired output.

Advantages of Batch Operating System

- ① It is useful when we working with larger files, which can take more time to execute.
- ② Multiple users can share batch systems.
- ③ The idle time for B.O.S is very less.
- ④ Easy to manage large work repeatedly in batch systems.

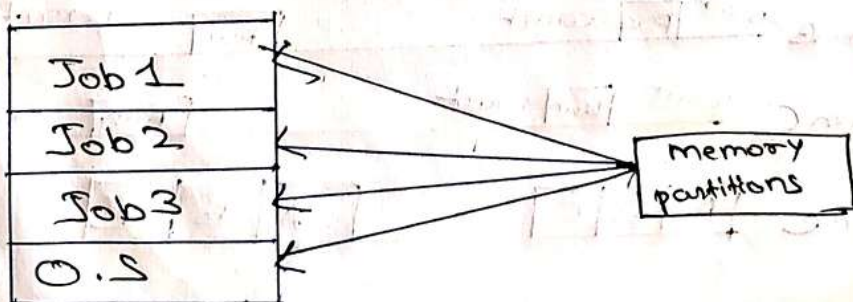
DisAdvantages

- ① They are hard to debug.
- ② It is sometimes costly.
- ③ Once the job is submitted user can't interact with it later.
- ④ The other jobs have to wait if any unknown job fails.
- ⑤ Difficult to predict time required for job.

Applications: Payroll Systems, Bank statements etc.

Multiprogrammed Operating System

Multiprogramming operating system can be simply illustrated as more than one program present in main memory and one of them kept for execution.
This is basically used for better execution of resources.



Non-Programming
The operating system keeps several jobs in memory at a time.

~~The operating systems keeps several job in memory at a time.~~

O.S. picks and begin to execute one of job in memory. Eventually the job may have to wait for I/O operation to complete.

In Non-Programming mode System; CPU will sit idle.

As long as there is always some job to execute, the CPU will never be idle.

Throughput of system will also increase.

throughput of system will also increase.

Program A:

Run

 wait

Run

 wait

a) Uni programming

Program A:

Run

 wait

Run

 wait wait

Program B: ~~wait~~

Run

 wait ~~wait~~

Run

 wait

Combined:

A	B
---	---

 wait

A	B
---	---

 wait

b) two (multiprogram)

3 programs

Program A:

Run

 wait

Run

 wait

Program B: wait

Run

 wait

Run

 wait

Program C: wait

Run

 wait

Run

 wait

Combine:

A	B	C
---	---	---

A	B	C
---	---	---

① $\subseteq$ A, B, C
 $A \subseteq (w) \subseteq \text{Run AB}$

Advantages of multi-Programming

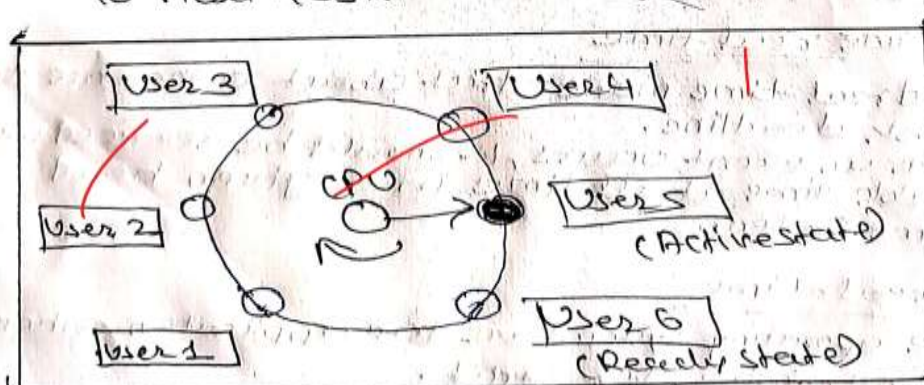
- * It provides less response time.
- * It may help to run various jobs in single application simultaneously.
- * Short-time jobs are done quickly in comparison to long time jobs.
- * Helps in reducing output time i.e. executes no. of program simultaneously by single processor.

DisAdvantages

- * There is expensive in
- * There is not any facility for user interaction of system resources with the system.
- * Multi programming system requires CPU scheduling
- * maintenance is expensive in multiprogramming.
- * It has a large no. of jobs, the long-term jobs will require long wait.

Time sharing Operating System

- Each task is given some time to execute so that all the tasks work smoothly.
- Each user gets the time of CPU as they use a single system.
- These system are also known as multitasking system.
- The task can be from a single user or different users also.
- The time that each task gets to execute is called quantum.
- After this time interval OS switches over to next task.



Advantages

- Each task get equal opportunity.
- Fewer chances of duplication of software.
- CPU idle time can be reduced.
- Avoids duplication of software.

Disadvantages

- Reliability problem.
- Data communication problem.
- One must take care of the security and integrity of user programs and data.

Multiprocessor System

1) When more than one CPU are used in execution of the processes it is known as multiprocessor system.

2) Many CPU are assigned in one cabinet to work in one cabinet for closed communication.

3) Multiprocessor system are also known as Parallel Systems or tightly coupled systems.

4) It is illustrated as multiprocessor which leads to better throughput rate.

5) Here CPU share Registers and cached memory and connected main memory.

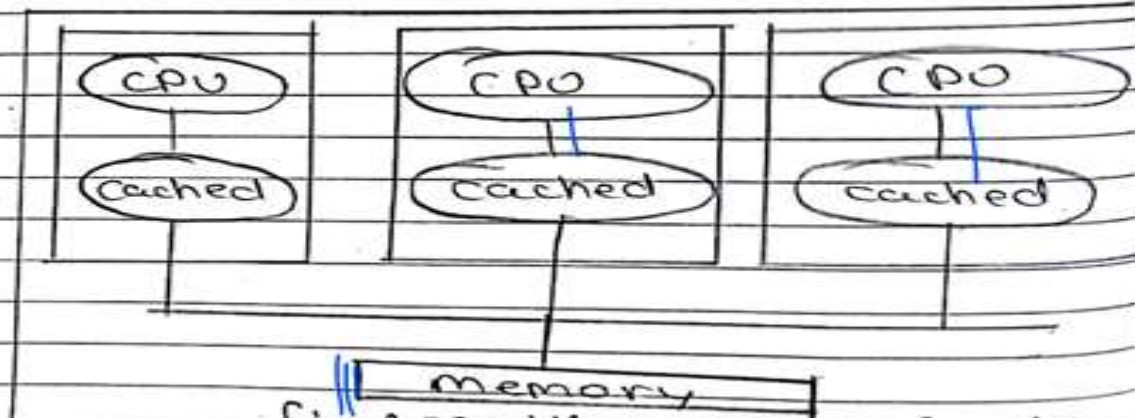


fig: multiprocessor system

There are two types of multiprocessor system.

1. Symmetric System
2. Asymmetric System.

1. Symmetric System

There is single multiple CPU which are treated equally.
No master-slave relationship.

2. Asymmetric System

There is one main CPU and other CPUs are followed or instructed by main. Follows Master-slave Relationship.

* Advantages

- 1) Faster response as process execute faster, increases response time.
- 2) Better throughput rate.
- 3) Parallel Systems: Processes can run parallelly at a time.
- 4) Faster execution, increases efficiency of system.
- 5) Cost Effective & multi

* Disadvantages.

- 1) Complex setup: Requires multiple processors (CPU) required cost, time.

P.T.O

- 2) Expensive: ^{many} more processor cost higher.
- 3) Can create a Deadlock Situation.
- 4) Maintenance is costly
- 5) Can create situation if process stops running should acknowledge to master node if

Distributed Operating System

Distributed systems use multiple central processors to serve multiple real-time applications and multiple users.

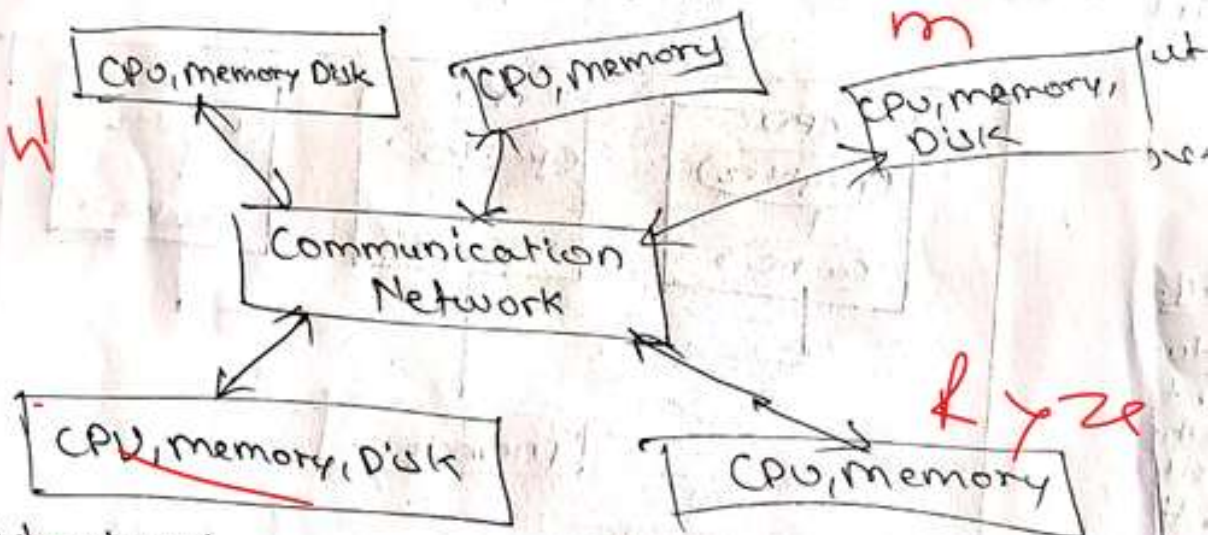
Data processing jobs are distributed among the processors accordingly.

The processors communicate with one another through various communication lines (such as high speed buses or telephone lines).

These are referred as loosely coupled systems or distributed systems.

Processors in a distributed system may vary in size and function.

These processors are referred as sites, nodes, computers, and so on.



Advantages

- Better services to customer.
- Reduction of delays in data processing.
- Reduction of load on the host computer.
- If one site fails in distributed system, the remaining sites can potentially continue operating.
- With resource sharing facilities, a user at one site may be able to use resources available at another.

Disadvantages

- Failure of main network will stop entire communication.
- To establish distributed systems the language is used not well-defined yet.
- These types of systems are not readily available as they are very expensive.

Example: Locus etc...

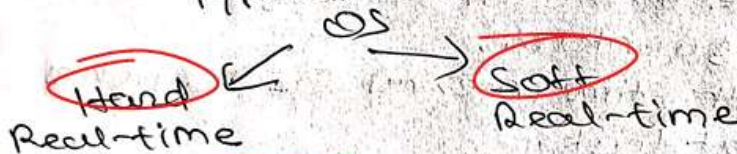
Real-time Operating Systems

These types of OS serve real time systems. The time interval required to process and respond to inputs are very small.

This time interval is called response time.

Real-time systems are used when there are time requirements that are very strict like missile systems, air traffic control, robots etc...

Types of Real-time



Hard real-time

Hard real-time means strict about adherence to each task deadline.

When an event occurs, it should be serviced within predictable time at all times in a given hard real time system.

Soft real-time

These operating systems are for application when time constraint is less strict.

Applications

- ↳ Flight Control Systems
- ↳ Simulations

Advantages

- ↳ maximize utilization of devices and system.
- ↳ focus on running applications over queued ones.
- ↳ designed to be error-free.
- ↳ efficient memory allocation.

Disadvantages

- ↳ supports only few simultaneous tasks.
- ↳ Can be resource-intensive.
- ↳ They are expensive.
- ↳ Algorithm can be complex and difficult to design.

Mobile OS

A mobile Operating System is one that controls smartphones, personal Digital Assistants (PDAs), tablet PCs, and information appliances.

A mobile operating system, also known as a OS, a mobile platform, or a handheld operating system, is the operating system that controls a mobile device.

Types of mobile OS

Android

iOS

Android

The Android Operating System is an open source operating system primarily used in mobile devices. It was written primarily in JAVA and based on the Linux operating system. It was initially developed by Android Inc and was eventually purchased by Google in 2005.

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iOS

iOS is the operating system created by Apple Inc. for mobile devices.

The iOS is used in many of mobile devices for apple such as iPhone, iPad, iPod etc.

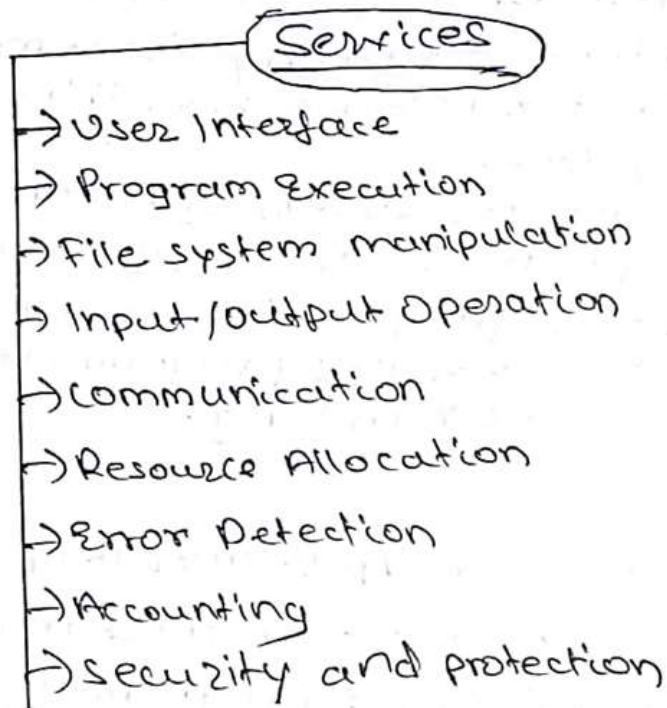
iOS is used a lot and only lags behind Android terms of popularity.

2.1: Different Services of Operating System.

An operating system provides different kinds of services to both users and to programs as well.

It also provides application programs (that run within an operating system) an environment to execute friendly

It provides users the services to run various programs in a convenient manner.



b) List of Operating System Services.

- 1) Accounting
- 2) User Interface.
- 3) Error Detection
- 4) Program Execution
- 5) File System Manipulation
- 6) Resource Management.
- 7) I/O device allocation
- 8) Security & Protection.

1) User Interface

- User can interact with the system using user-interface service of operating system.
- User interact with Operating System in three ways or interfaces:
 - 1) Command-line Interface
 - 2) Graphical User Interface
 - 3) Batch Based Interface.

2) Program Execution

- Computer or operating system have smaller main-memory and larger Secondary-memory.
- O/S can't access Secondary memory programs they need to bring in main-memory for execution.
- we not need to see allocation and deallocation O/S takes care of it.

I/O operations

I/O operations stands for Input/output Operations.

When program is in execution they may require some I/O devices or program may require files.

It is a responsibility of Operating System to allocate the I/O device or file to corresponding program.

For example, taking a printout, it is an duty of O.S to allocate printer to file.

User do
can't do permissions
directly

Scanning document, it duty of O.S to allocate scanner to file.

3) File System Manipulation.

- File System contains various directories & sub-directories which has file in it.
- We use manipulate the thing in File System or modification updation, deletion etc..... comes under this services. Such as creating a file, deleting a file, Renaming a file etc.....

5) Error Detection

→ Error is an abnormal condition occurs and system stuck.

→ It's duty or job of operating system if any error is discovered then inform user via interface.

Eg, mouse not working, system error etc...

6) Security & Protection

→ User can provide access to limited people in system is security.

→ Protection is the files should not get corrupted etc...

Communication

The Operating system manages the communication between processes.

One process can communicate with other process with the help of operating system.

May both process resides in same computer or may resides in other computer.

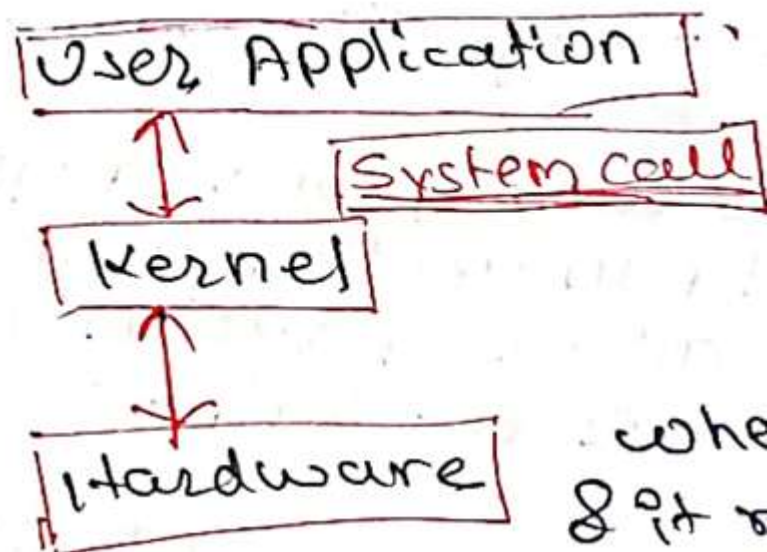
So, one process communicate with another process is known as interprocess communication.

Communication may be implemented by two method, either shared memory or message passing.

Communication between processes includes data transfer among them.

Accounting: Assume we have network, a collection of computers so multiple persons can work with computer.
multiple persons will be in network, so, which persons will use which resources, will be monitored by OS.
Example, P1 uses printer frequently
P2 uses scanner frequently

System calls



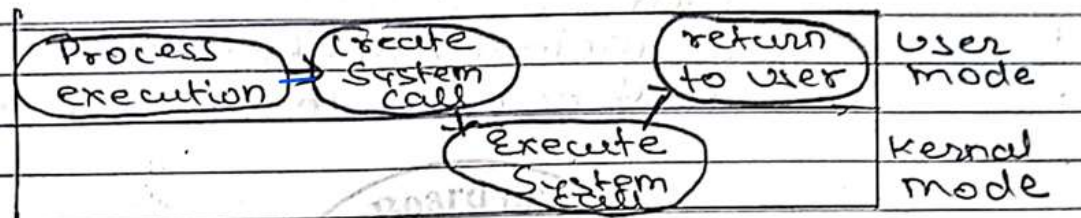
what is system call?

① The system call provides an interface to the Operating System services.

b) System call

System call is call that provide OS services to user via API (Application Programming Interface).

It is the interface between user and kernel.



Following are types of system call :-

- 1) Process control
- 2) File management
- 3) Device management
- 4) Communication
- 5) Information maintenance.

Types of System calls.

① Process & Job control

A running program needs to be able to halt its execution either normally (end) or abnormally (abort).

If program discovers an error in input & wants to terminate abnormally, it may also want to define error level.

When a process or job is running one program, it might need to start and run another program. ~~This is useful because it allows a system to follow instructions from a user or script to run specific tasks.~~

If we start a process or job we should be able to control its execution.

We may also want to terminate a job or process that we created (terminated process). If we find that it is incorrect or no longer needed we need waiting time to finish execution.

1) Examples of Process Control System call

2nd, abort

load, execute

→ create process, terminate process

→ wait for time, wait event, signal event

→ allocate and free memory

→ get process attributes, set process attributes.

② File Management

System calls which deal with operations related to files fall under this type.

System allows us to create and delete files.

For creating and deleting operating system call requires the name of file and other attribute of file.

File attributes such as file type, file size, protection codes etc. ---

System access these attributes for performing operation on file and directories.

Once file is created, we can open it and use it. System allows performing reading, writing or repositioning operation on file.

Examples → create file, delete file

→ open, close

→ read, write, reposition

→ get file attributes, set file attributes.

③ Device Management

When process in "running" state, it require several resources to execute. These resources include main memory, disk drivers, files and so on.

So system calls are used for asking permission from kernel. If resource is available, it is assigned to process.

Once the resource is allocated to process, process can read, write and reposition device.

→ Request device, release device

→ read, write, reposition

→ get device attributes, set device attributes

⑤ Communication

Process in system, communicate with each other.

Therpxpr (es)
Communication with each other is done by using two models: message passing & shared memory.

For transferring messages, sender process connects itself to receiving process by specifying receiving process name or identity.

Once the communication is over system close connection between communicating processes.

They need to communicate with each other for many reasons like if they need certain resources which is held by another process.

- create/delete communication connection
- send, receive messages
- transfer status information
- attach or detach remote devices

④ Information Maintenance

We need to keep information up to date so that system calls helps us to do that.

Transferring information between the user program and operating system requires system call.

System information includes displaying current date and time, the number of current users, the version number of operating system, the amount of free memory or disk space and so on.

Operating system keeps information about all its processes that can be accessed with system call such as get process attributes and set process attributes.

- get time or date, set time or date
- get system data, set system data
- get process/file or device attributes
- set process, file, or device attributes.

Question : State any two command line based operating systems and two GUI based operating system. (2marks)

ANSWER :

Command line based Operating System

1.DOS,2. UNIX

GUI based Operating System:

1. WINDOWS, 2.LINUX, 3.MaC OS

<u>Command Line Based OS</u>	<u>Graphical User Interface OS</u>
Interaction is by typing <u>commands</u>	Interaction with devices is by graphics and visual components and icons.
Commands need to be memorized	Visual indicators and icons are easy to understand
Less memory is required for storage.	More memory is required as visual components are involved
Accuracy is High	Accuracy is Low
CLI is difficult to use	GUI is easy to use
CLI is faster than GUI	GUI is slower than CLI
CLI operating system only needs keyboard.	While GUI needs both mouse & keyboard.
CLI appearance can't be changed or modified.	GUI appearance can be changed or modified
There are no graphics	graphics are used.
Does not avoid spelling mistakes and typing errors	Whereas GUI avoids type errors and spelling mistakes.
CLI do not use any pointing devices	While GUI uses pointing devices for selecting and choosing item.
In CLI we can obtain high precision	In GUI we can obtain low precision
Input is entered on cmd command prompt only.	While GUI, the input can be entered anywhere on the screen.

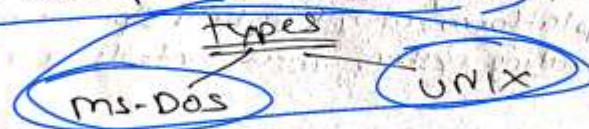
READ ONLY !!!

Command Line Based OS

The Command Line Interface interacts with the Operating System by typing commands/instructions on a command line.

Various commands need to be typed for carrying out various jobs like creating, deleting, renaming, or printing a file.

Usually in CLI each command represents an executable program, which is run when the command is typed with proper parameters.



DOS

DOS stands for Disk Operating Systems.

It was developed by Microsoft. So, Microsoft Disk Operating System.

It is a single user, non-graphical line oriented command or menu-driven operating system.

It has simple interface but interface is less user friendly.

Example: dir, mkdir etc.

(1981-2000)

dir
mkdir

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Command Line

Unix OS

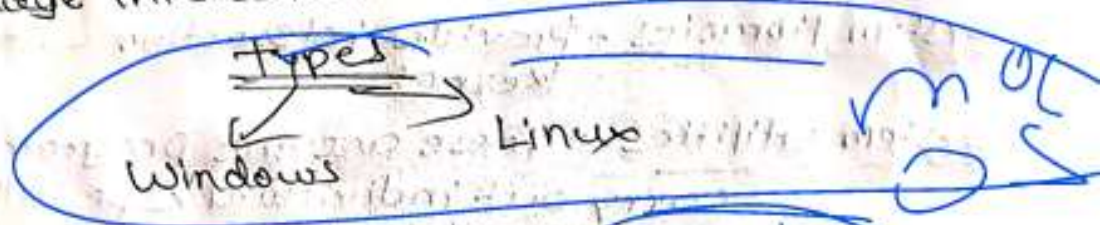
Developed in 1970 by Ken Thompson at Bell Laboratory.

Unix runs on a wider range of systems than any other O.S.

It allows people to work together and share information in controlled way, and flexibility.

GUI Based OS

A graphics-based operating system interface that uses icons, menus and a mouse (to click on the icon or pull down the menu) to manage interaction with system.



windows

- ↳ windows is developed by Microsoft.
- ↳ It is forerunner to the Apple Macintosh.
- ↳ Microsoft provided MS-DOS a graphical user interface called windows.

Components : ① Hardware Abstraction Layer
↳ provides portability across platforms.

② KERNEL

↳ It is responsible for handling all kind of program and process.

③ Executive

↳ It is written in C and includes 10 components.

Linux

- It is free and open source OS.
- It supports GUI
- It supports multiple languages.
- It supports multiple users, multithreading, multitasking & networking as well.
- It is portable and can be installed on any machine.

COMPONENTS OF OS

Q] List system component. Explain any two.

[S-15]

(List - 1 Mark, Description of any two - 1 ½ Marks each)

Q] List components of OS. Explain any one in detail.

[W-14]

(List - 1 Mark, Description of any one - 3 Marks)

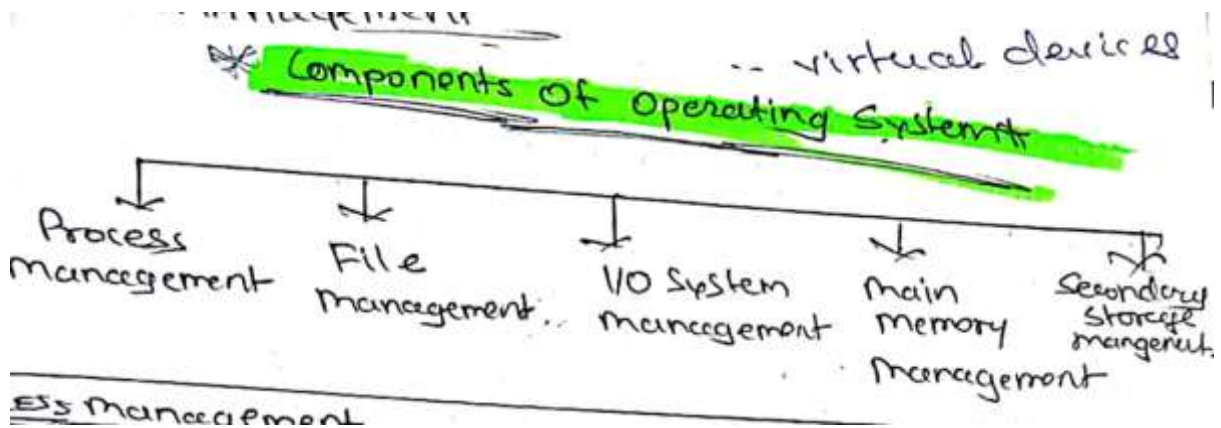
Q] Describe activities of I/O system and secondary storage management.

[S-17]

Q] List and explain components of operating system.

(6 MARKS)

[W-16]



Process management

A program is set of instructions.
When CPU is allocated to program it can start its execution.

A program in execution is process. The process which runs simultaneously are managed by process management.

Example, word processing system run by user on PC is a process.

A process needs various system resources CPU, time, memory, file and I/O devices to complete job execution.

These resources can be given to the process when it is created or allocated to it while its running.

The process management activity involves:-

- 1) Creation, execution and deletion of user and system processes.
- 2) A mechanism for deadlock handling.
- 3) A mechanism for process communication.
- 4) A mechanism for process synchronization.
- 5) Control the execution of user applications.
- 6) Suspension and resumption of processes.

Main-memory management
Main memory is an important resource of computer system that needs to be managed by operating system. To execute the program, user needs to keep the program in main memory.
Main memory is volatile.

So a program to be executed must be in the main memory. Services provided under Memory Management are to keep track of memory and allocating/ deallocating it to various processes. The OS keeps a list of free memory locations. Before a program is loaded in the memory from the disk, this MM consults the free list, allocates the memory to the process, depending upon the program size, updates the list of free memory.

The major activities of an operating in regard to memory-management are:

Keep track of which part of memory are currently being used and by whom.

Decide which **process** are **loaded** into memory when memory space becomes available

Allocate and de-allocate memory space as needed.

4. I/O System Management

Input / Output device management provides an environment for the better interaction between system and the I/O devices such as printers, scanners, tape drives etc.

To interact with I/O devices in an effective manner, the operating system uses some special programs known as device driver.

A device driver is a specific type of computer software that is developed to allow interaction with hardware devices.

Activities:

- Providing interfaces to other system components.
- Managing devices
- Transferring data
- Detecting I/O completion

File management

A file is a collected of related information defined by its creator.

Computers can store files on disk (secondary storage) which provide long term storage.

Some examples of storage media are magnetic tape, magnetic disk and optical disk.

Each of these media has its own properties like speed, capacity, and data transfer rate and access methods. A file system normally organized into the

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Directories ease their use.
These directories may contain file and other
directories.

Activities O.S responsible for file management

- Creation and deletion of files.
- Creation and deletion of directories.
- Mapping of file onto secondary storage.
- Backup of file on stable storage media.
- Support of primitive for management.

Secondary Storage Management

The computer system provides secondary storage to back up main memory.

Secondary storage is required because main memory is too small to accommodate all data and programs, and the data that it holds is lost when power is lost.

Most of programs including Compilers, Assemblers, word processors, editors etc. are stored on disk until loaded into memory.

Secondary storage consist of tapes, disks and other media.

O.S Responsible for following activities.

- ↳ Free space management
- ↳ Storage Allocation
- ↳ Disk scheduling.

List the activities performed in process management and file management (4marks)

b) Activities performed by process management.

- ① Provide a mechanism for Deadlock.
- ② Provide a mechanism for process synchronization.
- ③ Provide a mechanism for process monopolization.
- ④ Provide a mechanism for process scheduling.
- ⑤ Creating, deleting, termination process among system.
- ⑥ Divides processes into preemptive higher and lower priorities by preemptive scheduling.
- ⑦ Defines various process states.

Activities performed by File management.

- ① Free space management by techniques Bit map & Linked List.
- ② Creating File
- ③ Deleting a file
- ④ Write on a file
- ⑤ Read a file
- ⑥ Rename a file
- ⑦ Truncating the file
- ⑧ Repositioning a file
- ⑨ Accessing a file by - Random

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P.T.O

Scanned with CamScanner

Sequential, Swapping Method.

- ⑩ Allocating files by - Linked list allocation, contiguous allocation, index allocation.